



Security Monitoring & Wazuh SIEM Deployment

Overview

This document details the deployment and configuration of **Wazuh SIEM (Security Information and Event Management)** as the cornerstone of the Blue Team security monitoring infrastructure. The implementation establishes comprehensive security monitoring capabilities using an all-in-one Wazuh deployment on **Rocky Linux 9.6**, strategically positioned within the dedicated **BlueTeam VLAN**.

Deployment Objectives

Primary Goals

- Deploy latest **Wazuh SIEM (version 4.12.0)** for comprehensive security monitoring
- Establish centralized log collection and analysis capabilities
- Create foundation for threat detection and incident response
- Implement security monitoring without impacting other network segments

Strategic Placement

- VLAN Assignment:** BlueTeam VLAN (VLAN 20 - 192.168.20.0/24)
- Network Isolation:** Separated from other lab functions
- Security Focus:** Dedicated environment for defensive security operations

Server Preparation & Network Configuration

Hardware Specifications

Component	Details
Hostname	lab-core-rocky
Operating System	Rocky Linux 9.6 (Blue Onyx)
Architecture	x86_64
Network Interface	enp0s20f0u4
Switch Connection	Port 4 (Untagged VLAN 20)

Network Configuration Process

Initial Interface State Assessment

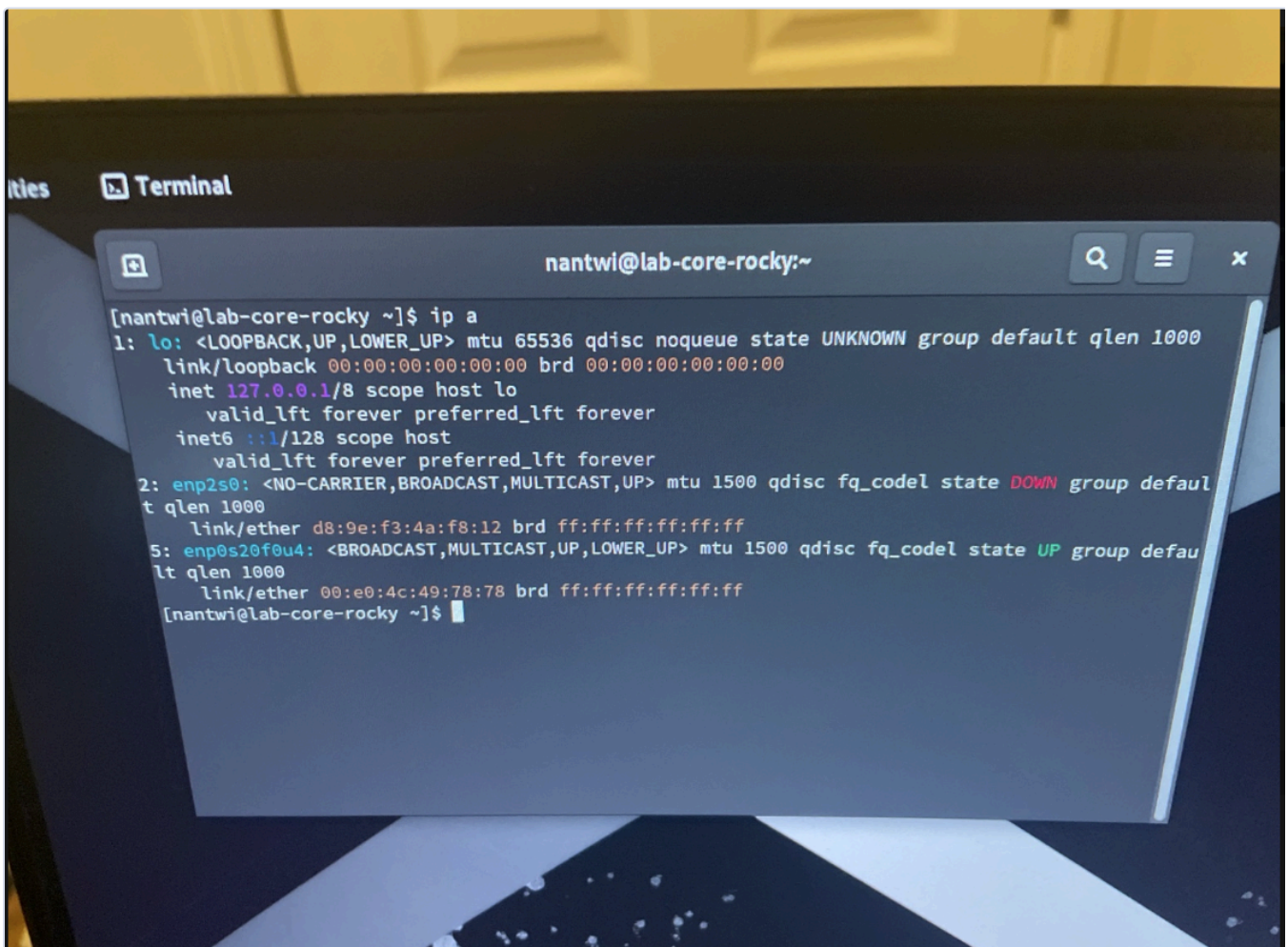
After connecting the server to **Port 4** and booting Rocky Linux:

Interface Status Check:

```
ip a
```

Observed State: - Interface `enp2s0` : **DOWN** (unused) - Interface `enp0s20f0u4` : **UP** (active, no IP assigned)

Network Setup Evidence



DHCP Verification & Testing

Before implementing static IP configuration, DHCP connectivity was verified to confirm proper VLAN assignment and network functionality.

DHCP Configuration Commands

```
# Create new network connection with DHCP
sudo nmcli con add type ethernet ifname enp0s20f0u4 con-name BlueTeam-DHCP ipv4.method auto

# Activate the connection
sudo nmcli con up BlueTeam-DHCP
```

DHCP Test Results

- ✓ **Dynamic IP Assignment:** 192.168.20.50/24 (within expected DHCP range)
- ✓ **Gateway Connectivity:** Successfully pinged 192.168.20.1
- ✓ **VLAN Verification:** Confirmed placement in BlueTeam VLAN

Static IP Implementation

For production SIEM deployment, a **predictable static IP address** was required to ensure consistent access and monitoring capabilities.

Target Static Configuration

Parameter	Value
IP Address	192.168.20.2/24
Gateway	192.168.20.1
DNS Servers	8.8.8.8
Connection Name	BlueTeam-Static

Static Configuration Commands

```
# Modify existing connection to static configuration
sudo nmcli con mod BlueTeam-DHCP \
  ipv4.method manual \
  ipv4.addresses 192.168.20.2/24 \
  ipv4.gateway 192.168.20.1 \
  ipv4.dns 8.8.8.8

# Restart connection to apply changes
sudo nmcli con down BlueTeam-DHCP
sudo nmcli con up BlueTeam-DHCP

# Rename connection for clarity
sudo nmcli con mod BlueTeam-DHCP connection.id BlueTeam-Static
```

Configuration Verification

```
[nantwi@lab-core-rocky ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether d8:9e:f3:4a:f8:12 brd ff:ff:ff:ff:ff:ff
5: enp0s20f0u4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:e0:4c:49:78:78 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.2/24 brd 192.168.20.255 scope global noprefixroute enp0s20f0u4
        valid_lft forever preferred_lft forever
    inet6 fe80::9d60:ee56:744a:6ef6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[nantwi@lab-core-rocky ~]$
```

Connectivity Validation

```
# Verify static IP assignment
ip a

# Test gateway connectivity
ping -c 4 192.168.20.1

# Verify internet connectivity
ping -c 4 8.8.8.8
```

Results:  All connectivity tests successful - static configuration operational

Wazuh SIEM Deployment Process

Pre-Installation System Preparation

System Updates & Dependencies

```
# Update system packages
sudo dnf update -y

# Install required utilities
sudo dnf install -y curl wget unzip gnupg2
```

Clean Previous Installations

Since Wazuh had been previously attempted on this system, comprehensive cleanup was performed to avoid conflicts:

```
# Remove any existing Wazuh packages
sudo dnf remove -y wazuh-* filebeat opensearch-dashboard

# Clean configuration and data directories
sudo rm -rf /etc/wazuh-* /var/lib/wazuh-* /usr/share/wazuh-* /etc/filebeat /var/lib/filebeat
```

Wazuh Installation Implementation

Installer Acquisition

```
# Download official Wazuh all-in-one installer
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh

# Make installer executable
chmod +x wazuh-install.sh
```

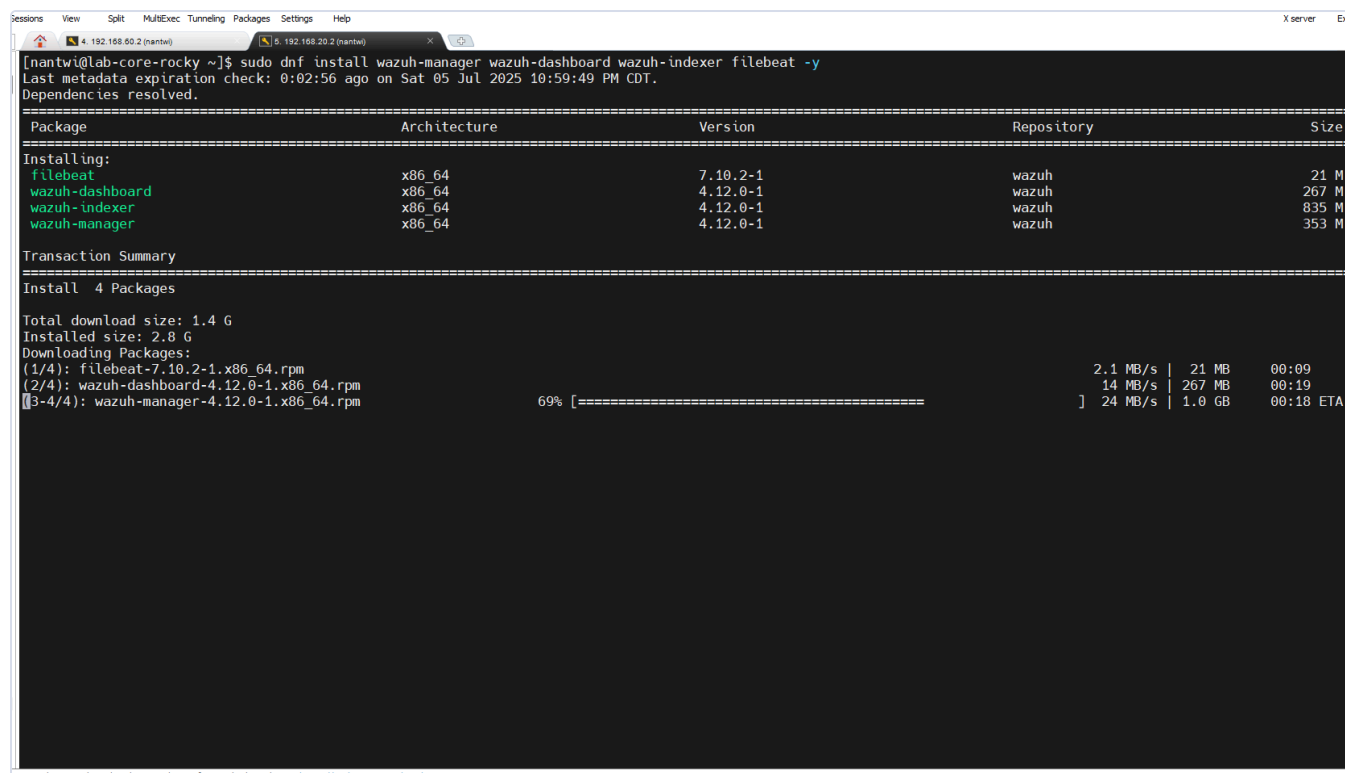
All-in-One Deployment

The Wazuh team's recommended deployment method was used for comprehensive SIEM functionality:

```
# Execute all-in-one installation
sudo ./wazuh-install.sh -a -i -o
```

Installation Parameters: - **-a** : Install all components (Manager, Indexer, Dashboard, Filebeat) - **-i** : Ignore OS compatibility check (Rocky Linux support) - **-o** : Remove old installations if present

Installation Process



```
[nantwi@lab-core-rocky ~]$ sudo dnf install wazuh-manager wazuh-dashboard wazuh-indexer filebeat -y
Last metadata expiration check: 0:02:56 ago on Sat 05 Jul 2025 10:59:49 PM CDT.
Dependencies resolved.

=====
Package                                Architecture    Version          Repository        Size
=====
Installing:
filebeat                               x86_64          7.10.2-1         wazuh             21 M
wazuh-dashboard                        x86_64          4.12.0-1         wazuh             267 M
wazuh-indexer                         x86_64          4.12.0-1         wazuh             835 M
wazuh-manager                         x86_64          4.12.0-1         wazuh             353 M
=====

Transaction Summary
-----
Install 4 Packages

Total download size: 1.4 G
Installed size: 2.8 G
Downloading Packages:
(1/4): filebeat-7.10.2-1.x86_64.rpm                2.1 MB/s | 21 MB   00:09
(2/4): wazuh-dashboard-4.12.0-1.x86_64.rpm         14 MB/s | 267 MB 00:19
[3-4/4]: wazuh-manager-4.12.0-1.x86_64.rpm         69% [=====] 24 MB/s | 1.0 GB 00:18 ETA
```

Installation Duration: Approximately 10-15 minutes

Components Installed: - **Wazuh Manager:** Core SIEM engine and rule processing - **Wazuh Indexer:** OpenSearch-based data storage and indexing - **Wazuh Dashboard:** Web-based management interface - **Filebeat:** Log shipping and data forwarding

Installation Completion

Upon successful installation, the system provided: - **Dashboard URL:** `https://192.168.20.2:443` - **Admin Credentials:** Automatically generated secure password - **SSL Certificates:** Automatically configured for HTTPS access

Post-Installation Configuration

Firewall Configuration

Rocky Linux ships with `firewalld` enabled by default. Essential Wazuh ports were opened for proper operation:

```
# Open Wazuh Manager ports
sudo firewall-cmd --permanent --add-port=1514/tcp # Agent communication
sudo firewall-cmd --permanent --add-port=1514/udp # Agent communication
sudo firewall-cmd --permanent --add-port=1515/tcp # Agent enrollment

# Open Dashboard port
sudo firewall-cmd --permanent --add-port=443/tcp # HTTPS web interface

# Apply firewall changes
sudo firewall-cmd --reload
```

Service Validation

Verification of all Wazuh components:

```
# Check Wazuh Manager status
sudo systemctl status wazuh-manager

# Check Wazuh Indexer status
sudo systemctl status wazuh-indexer

# Check Wazuh Dashboard status
sudo systemctl status wazuh-dashboard

# Check Filebeat status
sudo systemctl status filebeat
```

Service Status Verification

```
[nantwi@lab-core-rocky ~]$ sudo systemctl status wazuh-manager
sudo systemctl status wazuh-indexer
sudo systemctl status wazuh-dashboard
sudo systemctl status filebeat
[sudo] password for nantwi:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-07-06 00:13:22 CDT; 15h ago
     Tasks: 174 (limit: 99376)
    Memory: 6.3G
      CPU: 29min 59.795s
   CGroup: /system.slice/wazuh-manager.service
           └─106850 /var/ossec/framework/python/bin/python3 /var/ossec/ap/cripts/wazuh_apid.py
              106851 /var/ossec/framework/python/bin/python3 /var/ossec/ap/cripts/wazuh_apid.py
              106852 /var/ossec/framework/python/bin/python3 /var/ossec/ap/cripts/wazuh_apid.py
              106855 /var/ossec/framework/python/bin/python3 /var/ossec/ap/cripts/wazuh_apid.py
              106856 /var/ossec/framework/python/bin/python3 /var/ossec/ap/cripts/wazuh_apid.py
              106900 /var/ossec/bin/wazuh-authd
              106914 /var/ossec/bin/wazuh-db
              106940 /var/ossec/bin/wazuh-execd
              106952 /var/ossec/bin/wazuh-analysisd
              106982 /var/ossec/bin/wazuh-syscheckd
              107029 /var/ossec/bin/wazuh-remoted
              107068 /var/ossec/bin/wazuh-logcollector
              107085 /var/ossec/bin/wazuh-monitord
              107095 /var/ossec/bin/wazuh-modulesd
Jul 06 00:13:17 lab-core-rocky env[106788]: Started wazuh-analysisd...
Jul 06 00:13:18 lab-core-rocky env[106788]: Started wazuh-syscheckd...
Jul 06 00:13:19 lab-core-rocky env[106788]: Started wazuh-remoted...
Jul 06 00:13:19 lab-core-rocky env[106788]: Started wazuh-logcollector...
Jul 06 00:13:19 lab-core-rocky env[106788]: Started wazuh-monitord...
Jul 06 00:13:19 lab-core-rocky env[107093]: 2025/07/06 00:13:19 wazuh-modulesd:router: INFO: Loaded router module.
Jul 06 00:13:19 lab-core-rocky env[107093]: 2025/07/06 00:13:19 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Jul 06 00:13:20 lab-core-rocky env[106788]: Started wazuh-modulesd...
Jul 06 00:13:22 lab-core-rocky env[106788]: Completed.
Jul 06 00:13:22 lab-core-rocky systemd[1]: Started Wazuh manager.
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-07-06 00:09:18 CDT; 15h ago
     Docs: https://documentation.wazuh.com
    Main PID: 101633 (java)
      Tasks: 90 (limit: 99376)
     Memory: 1.5G
        CPU: 11min 546ms
   CGroup: /system.slice/wazuh-indexer.service
           └─101633 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: System:setSecurityManager has been called by org.opensearch.bootstrap.OpenSearch (file:/usr/share/wazuh-indexer/lib
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.OpenSearch
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: Jul 06, 2025 12:09:10 AM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: COMPAT locale provider will be removed in a future release
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: A terminally deprecated method in java.lang.System has been called
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: System:setSecurityManager has been called by org.opensearch.bootstrap.Security (file:/usr/share/wazuh-indexer/lib
Jul 06 00:09:10 lab-core-rocky systemd-entrypoint[101633]: WARNING: Please consider reporting this to the maintainers of org.opensearch.bootstrap.Security
Jul 06 00:09:10 lab-core-rocky systemd[1]: Started wazuh-indexer.
[lines 1-21/21 (END)] ...skipping...
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-07-06 00:09:18 CDT; 15h ago
     Docs: https://documentation.wazuh.com
    Main PID: 101633 (java)
      Tasks: 90 (limit: 99376)
     Memory: 1.5G
        CPU: 11min 546ms
```

Service Status:  All services active and running properly

Wazuh Dashboard Access & Verification

SSH Access to Wazuh Server

Remote access to the Wazuh server instance for administration and troubleshooting:

SSH Connection Details:

```
# SSH into Wazuh server
ssh nantwi@192.168.20.2
```

Parameter	Value
Hostname	lab-core-rocky
IP Address	192.168.20.2
SSH Port	22 (default)
SSH Username	nantwi
VLAN	BlueTeam (VLAN 20)

Dashboard Accessibility

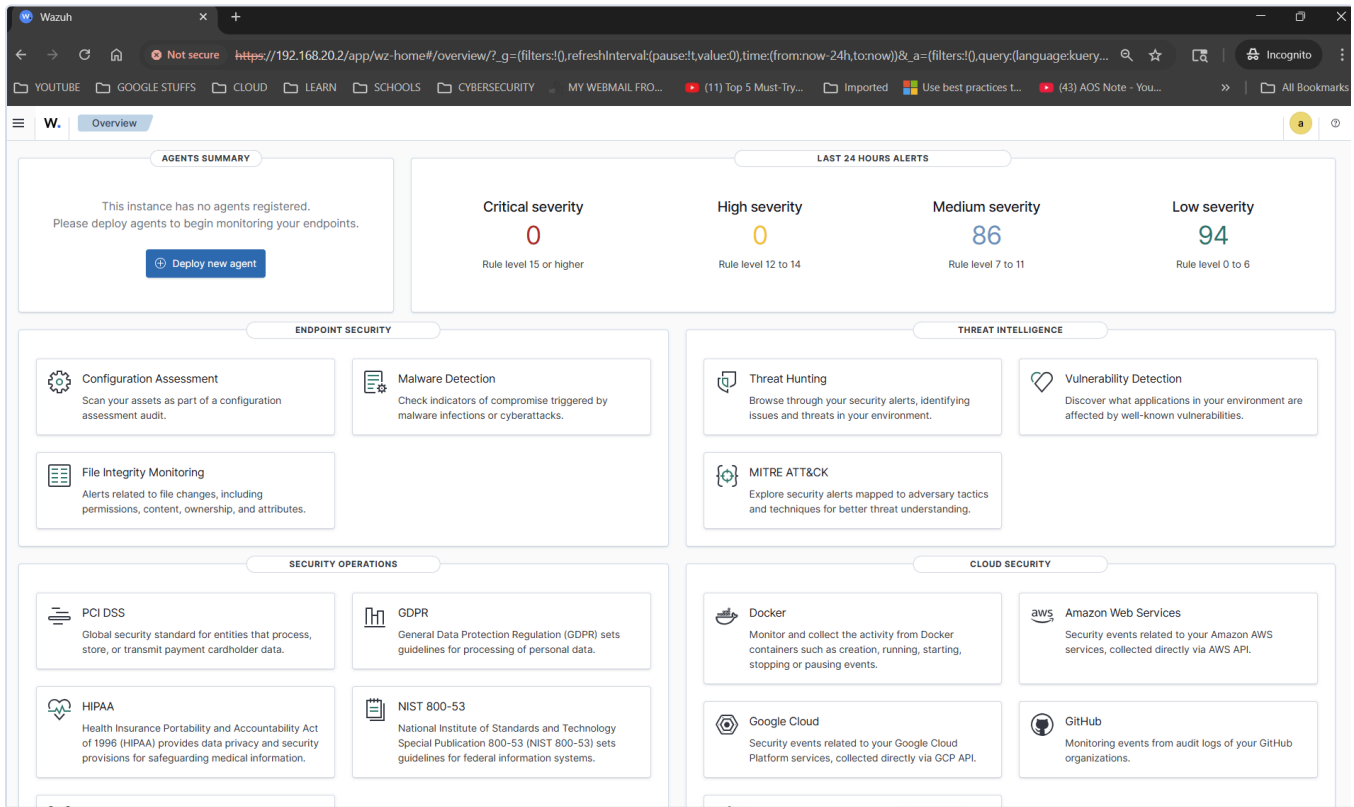
The Wazuh dashboard became accessible immediately after installation:

Access URL: <https://192.168.20.2>

Protocol: HTTPS with automatically generated SSL certificates

Authentication: Admin credentials provided during installation

Dashboard Interface



Wazuh Version Confirmation

- **Wazuh Version:** 4.12.0 (Latest stable release at deployment time)
- **Deployment Type:** All-in-One configuration
- **Components:** Manager, Indexer, Dashboard, Filebeat all operational

Initial Dashboard Features

- **Security Events:** Real-time security event monitoring
- **Agent Management:** Ready for endpoint agent deployment
- **Rule Management:** Pre-configured security rules and decoders
- **Compliance Dashboards:** Built-in compliance reporting
- **Threat Intelligence:** Integrated threat detection capabilities

Security Implementation

Network Security

- **VLAN Isolation:** SIEM contained within BlueTeam VLAN (192.168.20.0/24)
- **Firewall Protection:** pfSense controls all inter-VLAN communication

- **SSL Encryption:** Automatic HTTPS certificate generation and implementation
- **Access Control:** Dashboard access restricted to authenticated users
- **Port Security:** Only necessary ports opened in firewall configuration

Authentication & Access

- **Default Credentials:** Secure admin password automatically generated
- **HTTPS Enforcement:** All dashboard communication encrypted
- **Certificate Management:** SSL certificates automatically provisioned
- **Session Security:** Web-based authentication with session management

Operational Capabilities

Current SIEM Functionality

- ✓ **Real-time Monitoring:** Active security event collection and analysis
- ✓ **Log Management:** Centralized log aggregation and storage
- ✓ **Rule Engine:** Pre-configured security detection rules operational
- ✓ **Dashboard Interface:** Web-based management and monitoring
- ✓ **Agent Framework:** Ready for endpoint agent deployment

Built-in Security Features

- **Intrusion Detection:** File integrity monitoring capabilities
- **Vulnerability Assessment:** Integrated vulnerability scanning
- **Compliance Reporting:** PCI DSS, GDPR, HIPAA compliance frameworks
- **Threat Intelligence:** Integration with external threat feeds
- **Incident Response:** Automated response and alerting capabilities

Integration with Lab Infrastructure

Network Integration

- **VLAN Placement:** Strategic positioning in BlueTeam VLAN
- **Gateway Access:** Proper routing through pfSense (192.168.20.1)
- **Internet Connectivity:** Outbound access for threat intelligence updates
- **Inter-VLAN Security:** Controlled access from Management VLAN for administration

Administrative Access

- **Management VLAN Access:** pfSense rules allow administrative connectivity
- **SSH Access:** Available for system administration and maintenance
- **Web Dashboard:** HTTPS access for security operations
- **Remote Monitoring:** Integration prepared for Tailscale mesh VPN



Performance & Scalability

System Resources

- **CPU Utilization:** Optimized for all-in-one deployment
- **Memory Usage:** Sufficient resources allocated for SIEM operations
- **Storage Capacity:** Adequate space for log retention and indexing
- **Network Performance:** Full gigabit connectivity through managed switch

Scalability Considerations

- **Agent Capacity:** Supports hundreds of endpoint agents
- **Log Volume:** Configurable retention policies for storage management
- **Performance Tuning:** Options for optimizing based on log volume
- **High Availability:** Foundation prepared for future HA implementation



Deployment Validation

Functional Testing

- ✓ **Service Status:** All Wazuh components operational
- ✓ **Web Interface:** Dashboard accessible and responsive
- ✓ **Network Connectivity:** Full connectivity to gateway and internet
- ✓ **Firewall Rules:** Proper port access configured
- ✓ **SSL Certificates:** HTTPS encryption functional

Security Verification

- ✓ **VLAN Isolation:** Confirmed placement in BlueTeam VLAN
- ✓ **Access Controls:** Administrative access properly restricted
- ✓ **Service Security:** All services running with appropriate permissions
- ✓ **Network Security:** pfSense firewall rules effective



Implementation Notes

Rocky Linux Compatibility

- **Official Support:** Rocky Linux not officially listed by Wazuh
- **Practical Reality:** Installation and operation fully stable
- **Compatibility Flag:** `-i` flag used to bypass OS check
- **Production Viability:** No functional limitations observed

Installation Method

- **All-in-One Choice:** Selected for comprehensive functionality in single system

- **Alternative Options:** Distributed deployment available for larger environments
- **Resource Efficiency:** Single-server deployment optimal for lab environment
- **Future Migration:** Can expand to distributed architecture if needed

Maintenance & Operations

Regular Maintenance Tasks

- **System Updates:** Regular Rocky Linux security updates
- **Wazuh Updates:** Periodic SIEM platform updates
- **Rule Tuning:** Optimization of detection rules based on environment
- **Log Management:** Monitoring of storage usage and retention policies

Monitoring & Alerting

- **Service Health:** Monitor all Wazuh component status
- **Resource Usage:** Track CPU, memory, and storage utilization
- **Network Connectivity:** Ensure consistent network access
- **Security Events:** Regular review of security alerts and incidents

Next Phase Integration

Agent Deployment Planning

The SIEM infrastructure is now ready for: - **Linux Agent Deployment:** Ubuntu servers and workstations - **Windows Agent Integration:** Windows systems across lab environment - **Network Device Monitoring:** Integration with pfSense and switch logs - **Application Monitoring:** Custom log sources and application security

Security Operations Expansion

Prepared foundation for: - **Custom Rule Development:** Environment-specific detection rules - **Threat Hunting:** Proactive security investigation capabilities - **Incident Response:** Automated response and notification systems - **Compliance Reporting:** Regulatory compliance monitoring and reporting

Deployment Summary

Successfully Implemented

- **Wazuh SIEM 4.12.0:** Latest version deployed and operational
- **All-in-One Configuration:** Complete SIEM stack on single system
- **BlueTeam VLAN Integration:** Properly segmented and secured
- **Network Connectivity:** Full access with appropriate security controls
- **Web Dashboard:** HTTPS-secured administrative interface

Current Operational Status

Component	Status	Accessibility
Wazuh Manager	Active	Internal network
Wazuh Indexer	Active	Internal components
Wazuh Dashboard	Active	https://192.168.20.2
Filebeat	Active	Log shipping operational

Infrastructure Integration

- **Network Foundation:** Successfully integrated with pfSense VLAN architecture
- **Security Posture:** Enhanced with centralized security monitoring
- **Operational Readiness:** Prepared for agent deployment and rule customization
- **Administrative Access:** Secured administrative interface and SSH access

Security Monitoring Status: Complete and Operational
Next Phase: [Observability Stack Deployment](#)